Winter Sem 2023-24
IIT Kanpur
MTH204 for BS students

MTH204A (1)

1. If $\sim$ is an equivalence relation on a non-empty set $A$, then the set of equivalence classes of $\sim$ forms a partition of $A$. On the other hand, if $\{A_i\}_{i \in I}$ is a partition of a non-empty set $A$, then there is an equivalence relation on $A$ whose equivalence classes are precisely the sets $A_i$, $i \in I$.

2. Let $G$ be a group. Show that
   (i) identity element in $G$ is unique.
   (ii) For each $a \in G$, $a^{-1}$ is unique.
   (iii) $(a^{-1})^{-1} = a$
   (iv) $(a * b)^{-1} = b^{-1} * a^{-1}$ in $(G, *)$.

3. Define $(\mathbb{Z}/n\mathbb{Z})^{\times} = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z}, (a,n) = 1\}$

   Show that $(\mathbb{Z}/n\mathbb{Z})^{\times} = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} \mid \exists\, \bar{c} \in \mathbb{Z}/n\mathbb{Z}$ s.t. $\bar{a}\bar{c} = \bar{c}\bar{a} = \bar{1}\}$

   Define $\cdot$ on $(\mathbb{Z}/n\mathbb{Z})^{\times}$ by
   $\bar{a} \cdot \bar{b} = \overline{ab}$ for $\bar{a}, \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^{\times}$

   Show that $((\mathbb{Z}/n\mathbb{Z})^{\times}, \cdot)$ is a group.

4. Show that $(\mathbb{R}, +) \cong (\mathbb{R}_{>0}, \times)$

5. Show that any two cyclic groups of same order are isomorphic

6. Determine all the subgroups of $(\mathbb{Z}, +)$

7. Let $a, n$ are integers with $(a,n) = 1$ & $n > 0$. Prove

that $a^{\phi(n)} \equiv 1 \pmod{n}$

~~8. Show that [illegible] for every prime p~~

8 (i) Show that $x^p = x \ \forall x \in \mathbb{Z}/p\mathbb{Z}$

(ii) If $G$ is a finite group of prime order, then show that $G$ is cyclic

9. Classify groups of order $\leq 5$.

10. Prove that $a\mathbb{Z} + b\mathbb{Z} = \gcd(a,b)\,\mathbb{Z}$

$$a\mathbb{Z} \cap b\mathbb{Z} = \operatorname{lcm}(a,b)\,\mathbb{Z}$$

11. (i) Show that if every element in a group is its own inverse, then the group is abelian

(ii) If $(a \cdot b)^r = a^r b^r \ \forall a, b \in G$, then $G$ is abelian.

12. Show that rational numbers with odd denominators forms a group with respect to addition.

13. Prove that $S_n, D_n$ are non-abelian for $n \geq 3$.

14. If $G$ is a finite group, show that $\exists\, n \in \mathbb{N}$ s.t. $a^n = 1 \ \forall a \in G$.

15. If $G$ is a group of even order, prove that it has an element $a \neq 1$ s.t. $a^2 = 1$

16. (i) Recall the definition of $GL_n(\mathbb{Z}/p\mathbb{Z})$. Show that $GL_n(\mathbb{Z}/p\mathbb{Z})$ forms a group w.r.to matrix multiplication

(ii) Determine the order of the group $GL_n(\mathbb{Z}/p\mathbb{Z})$. (Here $p$ is a prime).

17. Let $G$ be a group s.t. the intersection of all of its subgps different from $\{1\}$ is a subgroup different from $\{1\}$. Prove that every element in $G$ has finite order.

18. (i) prove that every subgroup of a cyclic group

is cyclic.

(ii) Let $G$ be a finite cyclic group. For every $d \mid n$, show that $G$ has a <u>unique</u> subgroup of order $d$.

(iii) Show that $G$ in (ii) has $\phi(n)$ many generators.

<u>19</u>: Let $G$ be a cyclic group of order $n$. If $d \mid n$, show that no. of elements of order $d$ in $G = \phi(d)$

If $d \nmid n$ then no of elements of order $d = 0$.

20. If $N$ & $M$ are normal subgp of $G$, prove that $NM$ is also a normal subgroup of $G$.

21. If $N$ & $M$ are normal subgroups of $G$ & $N \cap M = \{1\}$ Then show that $\forall\, n \in N, \forall\, m \in M,\ nm = mn$.

22. If $N$ is the only subgroup of order $|N|$ (which is finite) in $G$, Then show that $N$ is a normal subgroup of $G$.

<u>23</u>. Show that any subgroup of index 2 is normal in $G$.

<u>24</u>. Show that $\langle -1 \rangle$ is a normal subgp in $Q_8$.

<u>25</u>. Show that $A_3$ is normal in $S_3$ & $S_3/A_3 \cong \mathbb{Z}/2\mathbb{Z}$. Show that none of the two cycles are normal in $S_3$.

<u>26</u>. Determine all groups of order 6.

<u>27</u> If a cyclic subgroup $T$ of $G$ is normal in $G$, show that every subgroup of $T$ is normal in $G$.

28. If $N$ is normal in $G$, prove that $o(\bar{a})$ in $\frac{G}{N}$ divides $o(a)$ in $G$.

<u>29</u>. If $N$ is normal in a finite group $G$ s.t $[G:N]$ & $|N|$ are relatively prime, then show that any element $x \in G$, satisfying $x^{|N|} = e$ must be in $N$.

30. Verify Isomorphism theorems I & III in $D_8$, $Q_8$ & $\mathbb{Z}/n\mathbb{Z}$, $\mathbb{Z}$ through various examples

31. Let $G$ be finite group. If $o(x)=n$, show that $o(x^a) = \frac{n}{\gcd(a,n)}$. If $G$ is abelian & has elements $x, y$ of order $m, n$ respectively then show that there exists an element $z \in G$ of order $= \operatorname{lcm}(m,n)$

32. Prove that (i) $\phi(p^r) = p^r - p^{r-1}$

(ii) $\phi(mn) = \phi(n)\,\phi(m)$ if $(m,n)=1$.

33. Let $G$ be a finite group & if $\exists$ an automorphism $f: G \to G$ s.t. $f(x) = x^{-1}$ for strictly more that 75% elements $x$ of $G$, then prove that $f(x) = x^{-1}\ \forall x \in G$ & $G$ is an abelian group.

34. Given an example of a non-abelian group which sends each of its precisely 75% elements to their inverses.

35. Compute the Centre of $D_n$, $Q_8$, $GL_n(\mathbb{R})$.

36. If $G/Z(G)$ is cyclic, then prove that $G$ is abelian.

37. Let $G$ be a group of order $2n$. Let us assume that half of the elements of $G$ are of order 2 & the other half forms a subgroup $H$ of order $n$ (& $H$ has no element of order 2). Prove that $H$ is of odd order & is an abelian subgroup of $G$.

38. Let $G$ be the group of polynomials $(\mathbb{R}[X])$ in one variable with coefficient in $\mathbb{R}$. Define the map $I$ by

$I(0) = 0$

& $I(p(x)) = \int p(x)\,dx$. Show that $I$ is a

group homomorphism & determine the kernel.



39. Prove that if a group contains exactly one element of order 2, then that element is in the centre of the group.

40 Let A, B are abelian groups. Show that there is a natural group structure on Hom(A, B), the set of groups homomorphism from A to B, which makes Hom(A,B) an abelian group.

41. Determine Hom$(\mathbb{Z},\mathbb{Z})$, Automorphism$(\mathbb{Z},\mathbb{Z})$ Hom$(\mathbb{Z}, A)$ where A is any abelian group.

42. Prove that any group of order $p^2$, with p - a prime is abelian.

43. Let G be a finite group of order n & r be an integer relatively prime to n, then show that $g \in G \Rightarrow g = x^r$ for some $x \in G$.

44. Show that $A_4$ has no subgroup of order 6.

45. Determine the kernel & Image of the ~~map~~ natural map $\phi: \mathbb{Z}/8\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$

46 Deduce from Isomorphism theorems that there is a 1-1 correspondence between the subgroups of G containing N & subgroup of $G/N$. More precisely, there is a bijection from the set of subgroups of G which contains N onto the subgroups of $G/N$. In particular, every subgroup of $G/N$ is of the form $A/N$ for some subgroup A of G containing N.

47 Explicitely identify $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ inside $A_4$.

48. If G is group of order n & p is the smallest prime dividing order of G, then any subgroup of index p is normal in G.

49. Prove that the size of a conjugacy class containing an element $|\,|G|$

where $G$ is a finite group & for $x \in G$

Congugacy class of $x = \{gxg^{-1} \mid g \in G\}$

50. Show that any Group of $p$-power order has nontrivial centre, where $p$ is a prime.

51. (i) Prove Cauchy's theorem I: If $G$ is a finite abelian group & $p \mid |G|$ then $G$ contains an element of order $p$.

(ii) Deduce that ~~general same~~ the same result holds for a general finite non-abelian group.

52. Show that if a group has no nontrivial subgroup, then $G$ is finite of prime order.